

WASTEWISE-ML · REQUIREMENTS

WasteWise-ml requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	WML-REQ-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/WasteWise-ml		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3 status from WML-CAL-001; decisions of WML-DDR-001 applied (R9, R10, R12, R14); R3 verification size flagged
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

WasteWise-ml requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be revised after co-design sessions (see WML-PRB-001). No model has been trained, and at TRL 3 none will be: the status column gives the position on paper from the calculation note WML-CAL-001 and says plainly where a requirement is not met. No target was relaxed or redefined by the 2026-09-25 decisions (WML-DDR-001); they fix the design choices that the status depends on.

The **reference use case** is one person at a sorting bench with a low-cost Android phone on a stand over one item at a time on a light mat (decided, D1), or holding the phone by hand on the floor, on a dry mixed stream, in daylight or a lit shed, with no network. The app uses tap-to-scan (decided, D5).

Table 1. Requirements, targets and status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later, on hold)	Status at TRL 3 (WML-CAL-001)
R1	Material class accuracy	Top-1 accuracy of 90 % or more on the seven classes in <code>ml/data/taxonomy.yaml</code> , on items the model does not abstain on, on the field evaluation set (R11)	Evaluation on the held-out field set	At risk: expected full-coverage class error 9.1 % with about 1,290 photos per class, on an assumed learning curve
R2	Plastic grade accuracy from a photo	90 % or more for PET bottles and trays, 80 % or more for other rigid HDPE and PP, on non-abstained items	Field set, ground truth from NIR (WasteWise Scan) or legible resin codes	Not met by design for opaque HDPE versus PP and for black plastics (about 9 % of items); these go to "Unsure" and to WasteWise Scan. At risk for PET
R3	Hazard flag	Recall of 98 % or more for batteries, sharps and chemical containers; a hazard result is never offered as a sortable grade	Field set with at least 200 hazard items (WML-CAL-001 shows 200 proves 98 % only with zero misses; 400 proposed, N1, awaiting Amish)	At risk: hazards are rare (about 300 in 10,000 natural photos against 2,400 needed)
R4	Abstain when unsure	Model says "Unsure" below a set confidence; coverage (share of items answered) 70 % or more while meeting R1	Coverage and accuracy curve on the field set	At risk: about 68 % answered (65 % with a grade, 3 % hazard flags), 32 % "Unsure"
R5	Offline, on-device speed	Result within 0.3 s of the photo on the reference phone (R12), with no network	Timing on the reference phone	Met on paper: 64 to 201 ms for MobileNetV3-Large in 8-bit form (decided, D2)
R6	Small download	Model file 10 MB or less; app with model 30 MB or less	File sizes of the exported models	Met on paper: 4.4 MB in 8-bit form; app 12 to 19 MB. Float32 (17.5 MB) would not meet it
R7	Quick to use	3 s or less from placing an item to seeing the result, including tap and camera focus	Timed trials with users	Not verifiable at TRL 3: time budget 0.9 s nominal, 2.0 s worst
R8	Usable without reading	Result shown as icon, color and bin number, with optional voice in the local language; users new to the app sort correctly after a 15 min briefing	Co-design sessions and usability trials	Not verifiable at TRL 3; depends on co-design
R9	Local grades	Buyer grade names and bin numbers set per site in a configuration file, without retraining, for grades the model already separates	Configuration review	Met by design review (per-site file decided, D4); new visual grades still need retraining
R10	Battery life	A full 8 h shift with up to 500 scans uses 30 % or less of a 4,000 mAh phone battery	Energy estimate; later measurement	Met on condition: 8 % with tap-to-scan and the screen sleeping between scans; 51 % with the screen on all shift (met only)

ID	Requirement	Target	Verification (TRL 4 or later, on hold)	Status at TRL 3 (WML-CAL-001)
				with the power bank); 130 % with a live preview
R11	Field evaluation set	2,000 or more verified images from at least two sites, 50 or more per grade, never used for training	Dataset manifest	Not met: the set does not exist. Sized at 2,000 (400 hazards, 900 non-hazard minimum, 700 natural mix), about 43 h of labeling
R12	Reference phone	Runs on an Android phone costing about \$150 or less, 3 GB RAM, 2019 or later chipset	Device test	Met on paper: \$130 indicative reference phone, one per pilot station (decided, D8)
R13	Open and licensed	Code and weights MIT; every dataset license in <code>ml/data/SOURCES.md</code> allows training and publishing weights; a model card with per-class results and known failure modes	License review	At risk: TrashNet (MIT) and TACO (annotations CC BY 4.0, images per image) usable with checks; ZeroWaste is CC BY-NC 4.0 and must stay out of training (N2, awaiting Amish)
R14	Consent and privacy	Field photos only with informed consent; no faces or identifiable people kept; upload opt-in; pickers' organization co-owns field data (decided, D6)	Data protocol review with the partner	Not verifiable at TRL 3: protocol to be written with the partner, who is not yet chosen

■ Requirements not met or at risk

- **R2 is not met by a camera alone** for opaque HDPE versus PP and for black plastics. The design routes these to "Unsure" and to WasteWise Scan.
- **R11 is not met:** there is no field evaluation set, so R1 to R4 cannot be verified. Building it is TRL 4 work, on hold.
- **R10 is met only if the screen sleeps between scans** (or with the power bank); tap-to-scan alone is not enough.
- **R1, R3, R4 and R13 are at risk:** public data is small, clean and not organized by grade; hazards are rare; coverage is estimated at 68 %; one candidate dataset has a non-commercial license.

■ Assumptions

- Seven material classes from the taxonomy (plastic, paper, metal, glass, organic, composite and hazardous), with 22 grades below them.
- A phone battery of 4,000 mAh at 3.85 V holds 15.4 Wh; screen, camera and processor draw about 2.5 W while scanning and the screen alone about 0.9 W (estimates, WML-CAL-001 section 5).
- Published MobileNetV3 speeds on a Pixel 1 large core are a fair lower bound; a low-cost 2019 or later phone is taken as 1 to 4 times slower (estimate).
- The stream composition, learning curve and error ranking in WML-CAL-001 sections 7 and 8 are assumptions to be replaced by field data.